METRIC

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28 September 1995
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29 January 1979

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, M795
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament, Research Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for Projectile, 155MM, HE, M795 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification

MIL-STD-453 - Inspection, Radiographic

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: DODSSP Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. Army Armaments, Research, Development and Engineering Center (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9312769 -	Projectile, 155MM, HE, M795 Loading, Assy & Marking
9329574 -	Packing and Marking for Pallet Wood Ammunition for Projectile, 155MM, HE, M795
12914619 -	Packing and Marking for 155MM Projectile Pallet

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

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3. REQUIREMENTS

3.1 Material. Material and parts shall be in accordance with applicable drawings and specifications.

3.1.1 Clean explosive scrap. Clean explosive scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweepings, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.6).

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9312769, all associated drawings, and with all requirements specified in applicable specifications and drawings.

3.3 Cavitation in high explosive charges.

3.3.1 Cracks and projected cavities. Cracks and cavities within the explosive charge shall not exceed the requirements specified in Table I when determined in accordance with 4.5.1.1. Cavities having a maximum (max) projected length of 0.8mm or less shall be disregarded, except in Segment A. Any defect in Segment A that is of sufficient size to register an indication on the radiograph shall be cause for rejection regardless of its projected area.

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TABLE I Cast Defects

	SEGMENTS (See Figure 1)			
	A	B	C	D
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings, square millimeters (sq. mm)	0	161.0	323.0	323.0
Projected length of any cavity excluding pipes, cracks, and annular rings (mm).	0	12.7	12.7	19.0
Piping cavities max. projected area (sq. mm).	0	161.0	323.0	-
Piping cavities max. projected width (mm)	0	6.4	6.4	-
Cracks, max. projected width (mm)	0	0.8	0.8	-
Annular rings, max, projected width (mm)	0	0	6.4	-

Piping cavitation is defined as cavitation located on or near the vertical center line of the projectile with the longitudinal axis (length) three times or greater than the width.

An annular ring is defined as cavitation that takes the form of a ring or portion thereof around the periphery of the cast.

3.3.2 Surface cavitation. Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. The max. dimension of any visible cavity in the faced surface of the charge shall not exceed 6.4mm in any direction and the total area of all cavities shall not exceed 80 percent of the surface area. (Cavities having a max. dimension of 0.8mm or less shall be disregarded in this computation as shall voids within the applicable forward region of the fuze well prescribed by dwg. 9312769).

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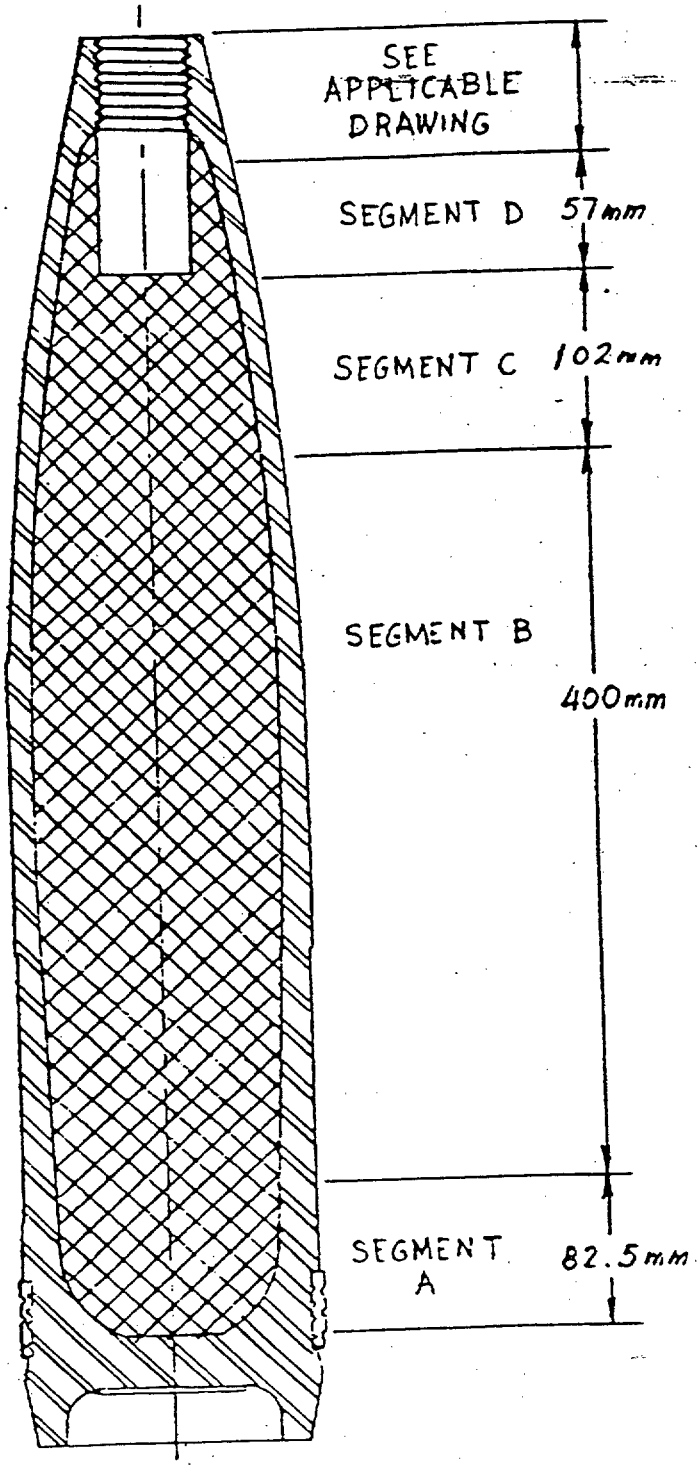


FIGURE 1. Projectile, 155mm, M795, HE Charge Segments

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3.3.3 Porous areas. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 Cracked charges. No transverse crack shall be permitted in Segment A and not more than a total of two transverse cracks shall be permitted in Segments B and C combined. Cracks shall be disregarded in Segment D.

3.3.5 Base separation. There shall be no discernible base separation between the base of the projectile cavity and the explosive charge when inspected by radiography. The maximum allowable base separation shall be .38mm (.015 in.) determined by sectioning, in accordance with 4.5.4.

3.3.6 Metal parts defects. There shall be no cracks, pipes or thin wall sections in the projectile.

3.3.7 Foreign material. There shall be no foreign material in the explosive charge.

3.4. Loading of high explosive.

3.4.1 Mating surfaces. The mating surfaces of threads shall be free of caked explosive and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care shall be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the projectile.

3.4.2 Solid explosives. Solid explosive shall not be in the projectile before loading or added to the projectile during initial pouring or cooling.

3.4.3 Coating. Projectile having the interior wall imperfectly coated shall not be loaded. The interior surface of the projectile shall be dry when the high explosive is poured into the projectile (see 4.4.2.1).

3.4.4 Projectile interior cavity. The interior cavity of the projectile prior to loading shall be free from foreign matter.

3.4.5 Fuze well. The fuze well shall be free from any foreign matter or loose explosive prior to assembly of the liner.

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3.4.6 Unsolidified pool of paint. There shall be no unsolidified pool of paint or blister in the projectile cavity prior to loading of the explosive.

3.4.7 Exterior surfaces. There shall be no explosive on the exterior surface of the projectile after loading.

3.4.8 Body cavity. The body cavity shall be free of scale, fins, burrs, draw marks, laminations, embedded foreign matter, folds, pits and sharp edges, in accordance with visual standards (see 6.11).

3.5 Proving ground. There shall be no premature burst or indication of metal parts separation in the gun bore or in flight.

3.6 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Exudation. After heat cycling of HE filled projectiles, there shall be no exudation in the fuze well or exterior of the projectile.

3.8 Workmanship. The requirements of workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.8.1 Cleaning. If required the cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.8.2 Protection of the rotating band. Care shall be exercised at all times so that the rotating band of each projectile will not be nicked or burred.

3.8.3 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.8.4 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust corrosion or other foreign matter, prior to inspection and shipping.

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3.8.5 Obturator. Care shall be taken at all times to protect obturator from being damaged.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual nor does it commit the Government to accept defective material.

4.1.2 General provision. Unless otherwise specified herein, the provisions of MIL-A-48078, apply and form a part of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Part Description	Drawing	Quantity
Projectile	9312769	24

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see Table II) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE II. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Projectile, 155mm, M795		See Below
			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH / INSPECTION METHOD REFERENCE
	Projectile, 155mm, HE, M795 Loading, Assy and Marking (Dwg. 9312769) Examination for Defects 1/	16, 100%	3.2 / 4.4.2.1 to 4.4.2.6 or 4.4.2.7 (as applicable)
	Proving Ground Test 2/	16, 100%	3.5 / 4.5.2
	Base Separation Special Radiographic inspection of sectioned projectile. After the test for base separation, each sample shall be completely sectioned and each half radiographed for the purpose of detailed examination of the explosive filler.	8, 100%	3.3.5 / 4.4.3.1.1 3.3 / 4.5.1.1. to 4.5.1.5 except that any convenient orientation may be used for radiography of sectioned projectiles

NOTES:
1. The examination for defects in paragraph 4.4.2 are for dimensional and visual inspections and in many cases must be witnessed during assembly.
2. After the dimensional and visual inspections of the 16 samples is completed subject these samples to the proving ground test.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. In addition, each inspection lot of loaded projectiles shall contain:

a. Projectile metal parts of one lot interfix number from one manufacturer.

b. Loaded projectile of one weight zone.

c. Fuze well liners from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification, and same specification revision.

d. TNT of the same type from one lot interfix number from one manufacturer.

4.4.2 Examinations and tests.

a. Classifications of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table III below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE III. Attributes sampling inspection.

Lot Size	Inspection Levels I	III	V
2 to 8	*	*	5
9 to 15	*	*	5
16 to 25	*	*	5
26 to 50	*	32	5
51 to 90	*	32	13
91 to 150	*	32	13
151 to 280	*	32	20
281 to 500	*	32	20
501 to 1200	*	80	20
1201 to 3200	1250	80	32
3201 to 10000	1250	125	32
10001 to 35000	1250	125	50
35001 to 150000	1250	125	50
150001 to 500000	1250	200	50
5000001 and over	1250	200	50

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.14). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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c. Automatic gaging. Facilities equipped with Government furnished automatic gaging capable of performing 100 percent inspection of any characteristic shall utilize this capability in accordance with contract.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Projectile Prior to Loading		9312769
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Special				
a	Foreign matter in projectile cavity	100%	3.4.4	Visual
b	Unsolidified pool of paint or blister in projectile cavity	100%	3.4.6	Visual/Manual
c	Interior metal defects	100%	3.4.8	Visual
Major 101	Interior protective coating with bare spot	Level III	3.4.3	Visual
Minor 201	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-63252A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Projectile and Liner (Prior to Assembly of Liner)	1 OF 1	9312769
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Presence of explosive on exterior of shell body, threads and forward face	100%	3.4.7	Visual
Major				
101	Depth of fuze well, max.	Level III	3.2	Gage
102	Foreign matter or loose explosive in fuze well	Level III	3.4.5	Visual
103	Projectile threads not clean prior to application of adhesive	Level III	3.2	Visual
104	Application of adhesive incomplete or incorrect	Level III	3.2	Visual
Minor				
201	Diameter of fuze well, maximum	Level V	3.2	Gage
202	Surface cavitation excessive	Level V	3.3.2	Visual
203	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-63252A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Projectile, prior to assembly of supplementary charge	1 OF 1	9312769
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Special a	Liner missing	100%	3.2	Visual
Major				
101	True position of inside diameter of liner, w/pitch diameter of threads	Level III	3.2	Gage
102	Depth to bottom of liner, min	Level III	3.2	Gage
103	Torque of liner (witnessed during application of assembly torque)	Level III	3.2	Visual
104	Swage missing or incomplete	100%	3.2	Gage
105	Inside diameter of liner after swaging	Level III	3.2	Visual
Minor				
201	Liner damage	Level V	3.2	Visual
202	Liner removable by hand	Level V	3.2	Manual
203	Foreign matter in liner cavity including explosive	Level V	3.2	Visual
204	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-63252A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Projectile, Prior to Assembly of Lifting Plug	1 OF 1	9312769
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Supplementary charge pad or supplementary charge missing	100%	3.2	Visual
Major				
101	Supplementary charge inverted	Level III	3.2	Visual
102	Supplementary charge not removable by hand	Level III	3.2	Manual
Minor				
201	Supplementary charge damaged	Level V	3.2	Visual
202	Foreign matter on threads including explosive	Level V	3.2	Visual
203	Corrosive preventive compound missing from threads	Level V	3.2	Visual
204	Spacer missing or damaged	Level V	3.2	Visual
205	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Projectile Assembly	1 OF 1	9312769
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Projectile zone weight incorrect	100%	3.2	Balance
2	Proving Ground Test 1/		3.5	4.5.2
Special				
a	Zone stenciling incorrect	100%	3.2	Visual
b	Obturator missing or damaged	100%	3.8.5	Visual
Major				
101	Diameter of bourrelet, max. (applies if bourrelet has been retouched)	100%	3.2	Gage
102	Torque test failure of lifting plug	Level III	3.2	Test
103	Lifting plug gasket missing	Level III	3.2	Visual
104	Any weight zone punch mark inadequate	Level III	3.2	Visual/Manual
Minor				
201	Grommet missing, damaged or improperly assembled	Level V	3.2	Visual
202	Rotating band damaged	Level V	3.8.2	Visual
203	Protective coating with total damaged area in excess of 161 square mm	Level V	3.8.3	Visual
204	Marking misleading or unidentifiable (except for zones), including incorrect color code	Level V	3.2	Visual
205	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:
1. See 4.4.3.2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-63252A (AR)

PARAGRAPH 4.4.2.6	TITLE Pallet (alternate)	SHEET 1 OF 1		DRAWING NUMBER 9329574
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pallet cover damaged or improperly assembled	Level III	3.2	Visual
102	DOD symbol missing or unidentifiable	Level III	3.2	Visual
103	Projectile of different zones on the same pallet	Level III	3.2	Visual
104	Strapping or wire missing or loose	Level III	3.2	Visual/Manual
Minor				
201	Number of projectile assemblies incorrect	Level V	3.2	Visual
202	Marking misleading or unidentifiable	Level V	3.2	Visual
203	Strapping or wire improperly engaged	Level V	3.2	Visual
204	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-63252A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Packing and Marking for 155MM Projectile Pallet	1 OF 1	12914619
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Improperly assembled or parts missing	Level III	3.2	Visual
102	Spacers loose or damaged	Level III	3.2	Visual
103	Rods not properly engaged in cover or base	Level III	3.2	Visual
104	Lifting plug of projectile missing or damaged	Level III	3.2	Visual
105	Improper torque on nut at top of rod	Level III	3.2	Gage
106	Cotter pin not properly engaged	Level III	3.2	Visual/Manual
107	Projectile of different zones on the same pallet	Level III	3.2	Visual
Minor				
201	Any marking missing incorrect or unidentifiable	Level V	3.2	Visual
202	Number of projectile assemblies	Level V	3.2	Visual
203	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables and the following:

4.4.3.1 Cavitation (see 3.3, except 3.3.2). Cavitation defects Metal parts defects and foreign material defects are as follows:

Critical:

1. Any cavitation, porosity, cracks, piping, or annular rings in Segment A .....Radiography
2. Cavitation, porosity, cracks, piping or annular rings, in segment B or C in excess of that permitted in Table I .....Radiography
3. Cracks in excess of that permitted in 3.3.4 .....Radiography
4. Base separation (see 4.4.3.1.1) .....Radiography
5. Cavitation in Segment D in excess of that permitted in Table I.....Radiography
6. Metal parts defects in excess of that permitted by 3.3.6 (see Section 6.9).
7. Foreign material in excess of that permitted by 3.3.7.

This test shall be performed 100 percent on all projectile assemblies. Each assembly shall be subject to radiography examination. The test shall be performed as specified in 4.5.1 using equipment in accordance with 4.4.4. Any assembly which has any of the above critical defects shall be classed defective and removed from the lot.

Projectiles rejected by radiographic examination may be drilled out to one half the depth of the "B" segment in Figure 1 and repoured, but not more than one drilling operation may be performed per warhead.

4.4.3.1.1 Base separation. Projectiles showing the largest base separation as determined by Radiographic examination prior to post cycle heat treatment shall be selected for sectioning at the following rate. Select projectiles at the rate of one (1) out of seven hundred (700) projectiles to be included in the post cycle heat treatment lot with a minimum of eight (8) samples for the first article and production lots until 3 consecutive production lots have passed the initial post cycle heat treatment base separation inspection, at which time the minimum sample size will reduce to two (2) samples per lot. When a production lot has been

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rejected for base separation the sample size will revert back to the minimum eight (8) samples. These projectiles are to be included in the post cycle heat treatment and section for determination of base separation thereafter. If insufficient samples are available with base separation based on radiographs, a sufficient number of samples shall be randomly selected from the post cycle heat treatment lot to make up the required sample. The samples selected for sectioning shall be placed in the coolest area of the post cycle igloo. Any projectile after sectioning which has base separation in excess of that permitted by the applicable requirement, shall be classed defective, and the entire post cycle heat treatment lot rejected. The test shall be performed as specified in 4.5.4 using equipment in accordance with 4.4.4. Any lot which fails after being post cycle heat treated three times shall be rejected.

4.4.3.1.2 Metal parts defects. (Critical defect) All radiographs of the projectile section loading assemblies shall be examined for evidence of cracks, pipes and visually apparent thin wall sections in the projectile. The presence of any such indication shall cause that assembly to be classed defective and removed from the lot (see section 6.9).

4.4.3.1.3 Foreign material. (Critical defect) All radiographs or projectile section loading assemblies shall be examined for foreign material in the explosive charge. The presence of any identifiable foreign material shall cause that assembly to be classed defective and removed from the lot.

4.4.3.2 Proving ground. The test shall be conducted in accordance with 4.5.2. If any projectile fails to comply with the applicable requirement the lot shall be rejected.

4.4.3.2.1 First article and initial production lot samples. Beginning with the first lot produced and continuing until three consecutive lots have complied with all the acceptance criteria contained herein 16 sample projectiles shall be selected at random from each lot for this test. If any critical defect occurs, the lot shall be rejected.

4.4.3.2.2 Production lot samples. After three consecutive lots have complied with 4.4.3.2.1, 8 sample projectiles shall be selected at random for this test. If any critical defect occurs, the lot shall be rejected. In

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the event that two successive lots fail to meet all the requirements of this paragraph, the provisions of 4.4.3.2.1 shall again apply until three successive lots meet the requirements specified therein.

4.4.3.2.3 Test validity. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be considered invalid and a new test shall be performed with additional samples.

4.4.3.3 Exudation. Minor defect. Eight projectiles from the hottest area of the post cycle oven shall be used for the sample. Any projectile with exudation shall be cause for rejection of the post cycle treatment batch. Inspection shall be performed per 4.5.3.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

4.5 Method of inspection.

4.5.1 Radiographic examination.

4.5.1.1 Determination of projectile cavitation. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the projectile and tangent to the base of its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of explosive under inspection, with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the shell. Such screening should be free of any radiographically observable features in the projected area under inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453 (see 6.8). In addition, the sensitivity of the radiographic technique shall

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be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.25 at every point within the projected area of explosive, including the base, with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.5.1.2 and 4.5.1.3. All defect-like features in the area of interest on a radiographic image shall be treated as defects unless identified as extraneous artifacts or metal parts irregularities. In the event an original radiograph leaves doubt as to a shell's acceptability because of difficulty in evaluating the size, location, or classification of a defect, the projectile shall be rotated about its longitudinal axis from the original exposure, and additional radiographs taken for clarification. Acceptability in all cases shall be judged on the basis of the film giving the largest projection, except in the case where a re-radiograph reveals multiple superimposed cavities, each of which individually complies with the requirements. All additional radiographs of a projectile for defect conditions will be identified on the film as retakes and stored; traceable to the original.

4.5.1.2 Check of sensitivity of radiographic technique. The sensitivity of radiographic technique shall be determined by means of penetrameters fabricated as specified herein and through the use of standards containing known defects. The penetrameters shall be placed on the radiographic source side of a projectile so that the 2T holes are at the point shown in Figure 2. Three penetrameters for each projectile shall be used. When radiographing a number of similar projectiles the frequency of the sensitivity check shall be no less than one in each fifty radiographs, and on a rotating ring, the penetrameter shall be used at one station for each complete rotation of the ring. The use of standards will be limited to the initial qualification of the radiographic technique at the start of production and at intervals stated in 4.5.1.4 provided that uniform results can be expected during the interval.

The sensitivity of the radiographic technique shall be considered satisfactory when the image of the penetrameters, including the 2T holes, are perceptible in accordance with MIL-STD-453 and when the required defect standards are perceptible.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8. The penetrameters shall be located as shown on Figure 2. For further details on

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construction of the penetrameter see MIL-STD-453. The standard projectile will be provided by the Government.

4.5.1.3 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units. Accuracy of densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurement on the radiographic film. Calibration shall be made at a step tablet density level within .5D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) or no more than 6mm in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by the Government. Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.1.4 Review of radiographic sensitivity and techniques (see 6.8). The sensitivity of the radiographic technique shall be reviewed by submitting at least one radiograph of the standard and of the projectile containing the penetrameter and a copy of radiographic procedures prior to the start of initial production. These radiographs shall be taken and processed using the same techniques as will be used in production. In addition one radiograph of the projectile standard and the penetrameter projectile shall be submitted for review once a month for the first three months of each contract and every three months thereafter. After qualification of the radiographic technique no changes to developing process, film, manufacturer or type radiographic source or exposure time shall be made without first submitting additional sample radiographs of the standard and penetrameter projectile for approval.

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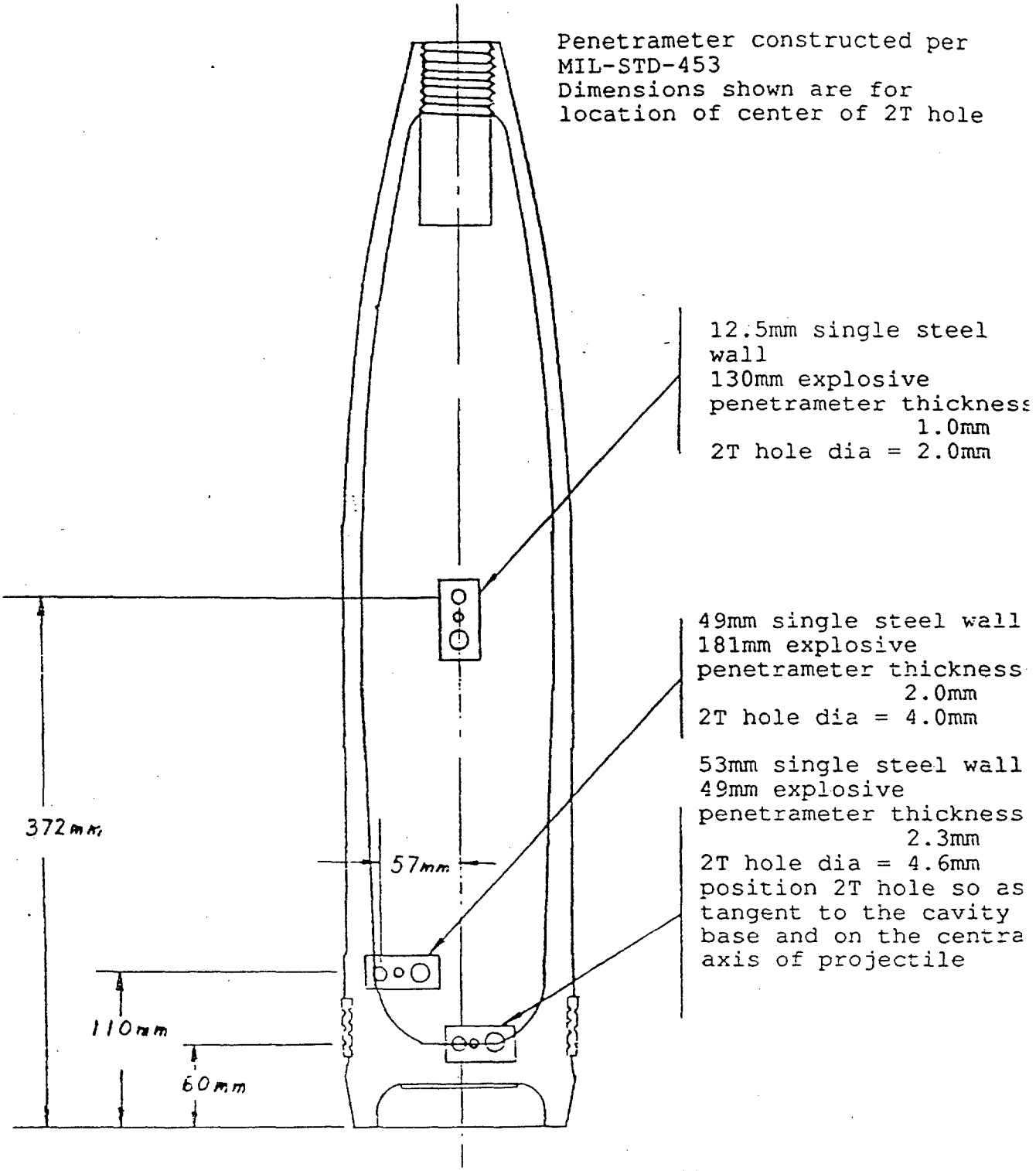


FIGURE 2. Projectile, 155mm, HE, M795
Radiographic Penetrameter Requirements

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4.5.1.5 Radiographic film. Only a film type and vendor as listed herein has been tested and approved by the Government for use in radiographing this projectile. The final choice of film type must be determined for the particular radiographic facility to be used and will be based on achieving adequate sensitivity on both standard and penetrameter projectile. A substitute item shall not be used before prior testing and approval by the Goverment (see 6.8.1).

4.5.1.6 Operator qualification. For operation of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I, or equivalent commercial standard as approved by the contracting officer. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

4.5.2 Proving ground test. This test shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The samples selected in accordance with 4.4.3.2 and Table II shall be fired in a M199 series cannon. The sample projectiles shall be assembled with a dummy or inert loaded fuze. Radar observation shall be made for muzzle velocity and range and deflection of impact (see. 6.12.1). The test results are to be recorded for individual items. Camera coverage shall be provided to observe the projectile in flight in order to record any malfunctions (see 6.12). In addition, radar shall be used to observe the projectile in flight in order to record any possible malfunctions.

4.5.2.1 First article and initial production lot samples. The 16 sample projectiles shall be divided into two equal groups. Group 1 shall be conditioned and fired at -54 degrees celsius at charge 8, M203 series propelling charge conditioned to -54 degrees celsius. Group 2 shall be conditioned and fired at 63 degrees celsius with a charge 8, M203 propelling charge conditioned to 63 degrees celsius.

4.5.2.2 Production lot samples. The 8 sample projectiles shall be conditioned and fired as Group 2, 63 degrees celsius with a charge 8, M203 propelling charge conditioned at 63 degrees celsius.

4.5.3 Exudation. The exterior of the projectile shall be visually inspected for exudation. If no exudate is found remove the lifting plug and inspect for exudation visually.

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4.5.4 Determination of base separation. After heat treatment the samples from each post cycle heat treatment lot shall be cooled to ambient temperature. The projectiles shall then be conditioned for a minimum of eight (8) hours at 21 ± 5 degree Celsius and sectioned at 21 ± 5 degree Celsius. A notch shall be cut from the base in the following manner: Cut parallel to the base approximately 90mm up from the base to the center of the projectile axis, next cut through the base toward the nose perpendicular to aforementioned cut so as to remove a wedge of steel to expose explosive and metal parts interface. Next subject projectile to an acceleration level of 200 to 250 G's in the nose down orientation and maintain projectile in that orientation to measure for base separation with a 1.25cm (max) wide feeler gage (see 6.13).

5. PACKAGING

5.1 Packing.

5.1.1 Packing - Level A. The projectiles shall be palletized in accordance with drawing 12914619.

5.1.2 Alternative Packing - Level A. Only with the specific approval of the contracting officer, the projectiles shall be palletized in accordance with drawing 9329574.

5.2 Marking. Markings shall be in accordance with drawing 12914619 or 9329574 as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The item covered by this specification is intended for use on the 155MM, Howitzer.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

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c. Requirements for submission of first article sample.

d. Applicable stock number.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armaments, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.6 Clean explosive scrap. Clean explosive scrap is defined as scrap formed in kettles, pails, risers, and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation.

6.7 Photographic densities. "American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York has been found satisfactory as a source of information for filter densities.

6.8 Radiographic procedures. Radiographic procedures, standards and sample radiographs shall be submitted to: Commander, ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, New Jersey 07806-5000 for review and approval.

6.8.1 Radiographic film sources. Approved list of film types and vendors for use in radiographing this projectile:

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Eastman Kodak Company
1187 Ridge Road, West
Rochester, New York 14650

Film Types: AA, T, AX-1 or M

E. I. DuPont de Nemours and Company
Photo Products Department
Wilmington, Delaware 19898

Film Types: NDT-55, NDT-65 or NDT-70

GAF Corporation
140 West 51st Street
New York City, New York 10020

Film Types: 800 or 400

6.9 Metal parts defects. The procuring activity will be notified of any projectile rejected for metal parts defects found upon inspection of the radiographs. The rejected projectile will be held pending disposition by the procuring activity.

6.10 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.11 Visual standards. Visual standards consistent with those used for metal parts acceptance shall be established at or provided to the contractor's plant in compliance with 3.4.8.

6.12 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to Commander, ARDEC, ATTN: AMSTA-AR-QAA-R, and AMSTA-AR-FSA-M, Picatinny Arsenal, NJ 07806-5000, and AMSMC-QAD-I Army Master Depository, Rock Island, IL 61299-6000.

6.12.1 Range and deflection. Range and deflection are not technical requirements of this specification. They are mandatory performance requirements that are achieved by

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design and are affected by variability in the manufacturing process that may not be completely controlled by the present TDP.

6.13 Projectile G loads. The 200 to 250 G load acceleration is applied to simulate projectile ramming. This is based on M107/M110 self-propelled artillery loader-rammer review program dated June 1971.

6.14 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.15 Qualification of post cyclic heat ovens. Before the post cyclic heat ovens are released for use in Loading Projectile, 155MM HE, M795 the following minimum test shall be run. Segregate by X-ray examination at least twenty-four (24) projectiles with indications of base separation (base separation is defined as air space between the shell metal part and the base of the HE fill). Holding six (6) samples as control rounds, heat cycle the remaining eighteen (18) by placing them through the normal cycle as it would be conducted during production taking care that the sample projectiles are located at the point in the ovens farthest from the incoming heating ducts. A schematic of the oven temperature distributions representing projectile exposure shall be submitted and the areas in which the control samples are placed shall be delineated. Approval of this facility is applicable to no more than the number of projectiles utilized in this qualification plus five percent. All projectiles should be inspected for base separation in accordance with 4.5.4. Record results being careful to clearly identify the control and test rounds. Forward results to Commander, ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, New Jersey 07806-5000.

6.16 Subject term (key word) listing.
Ammunition
Ballistic testing(security-temp. conditioned)
Base separation
Loading of High Explosive
Metric
Radiographic Inspection
Threaded Liner

6.17 Changes from previous issue. Marginal notations are not used in this revision to identify changes with

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respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project-1320-0007)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-63252A	28 SEPTEMBER 1995

3. DOCUMENT TITLE: PROJECTILE, 155MM, HE, M795 LOADING, ASSEMBLING AND PACKING

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable) 7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
COMMANDER US ARMY ARDEC	(1) Commercial: (201) 724-6628 (2) AUTOVON: 880-6628
c. ADDRESS (Include Zip Code) ATTN: AMSTA-AR-EDE-S/BLDG 12 PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — *Previous editions are obsolete.* — 198/290

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